



General Concept — FULL MIND MAP

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Chapter 01_i_General_Concept

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Big Bang Theory — Universe ki utpatti

- **Big Bang Theory** universe ki origin/evolution explain karta hai — ~13.8 billion years pehle ek hot, dense state se shuru hua. [Q1, Q2]
- Georges Lemaître expanding-universe idea ke pioneer the; cosmic expansion aur CMB isko support karte hain. [Q1]

Continental Drift Theory

- **A. Wegener (Alfred Wegener)** ne continental drift theory develop ki — German geophysicist. [Q3]



EXAM TRAP

- Continental Drift = **Wegener** (options mein Du Toit, Hess bhi distractors hote hain — Wegener hi correct hai) [Q3]

Milky Way aur Galaxy

- **Milky Way** = Spiral Galaxy (barred spiral); hamara Solar System iske Orion Spur/Arm mein hai. [Q4]
- Sun ko galactic centre ki 1 revolution karne mein ~22.5–25 crore years lagte hain — isko **Cosmic Year** / **Galactic Year** kehte hain. [Q5]
- Tare ek **particular shape ke group** ko **Constellation** (तारामंडल) kehte hain. [Q19]
- Total **88 constellations** officially recognized hain (IAU ke according). [Q16]

Stars — Colour, Temperature, Distance

- Star ka **colour uske temperature** ka indicator hota hai (Blue = hottest ~50,000K, Red = coolest ~2,000-3,500K). [Q6]
- **Sun** Earth ka sabse nearest star hai (~149.6 million km); Alpha Centauri Sun ke baad nearest star system hai; Sirius ~8.6 light-years door. [Q9]
- Betelgeuse — Orion constellation ka 2nd-brightest star; Hubble Space Telescope ne Sun ke alawa isi ka first direct image liya tha. [Q20]

Chandrasekhar Limit, Black Holes, Supernova

- **Chandrasekhar Limit** — S. Chandrasekhar (Indo-American physicist) ne bataya ki white dwarf ka max mass ~1.4 times solar mass hota hai; iske aage star collapse ho jaata hai (neutron star/black hole). [Q7, Q10, Q11, Q12]
- Chandrasekhar ko is work ke liye **1983 Nobel Prize in Physics** mila tha. [Q7, Q12]
- **Black Hole** = collapsed star, extremely high density; itna strong gravity ki light bhi escape nahi kar sakti — telescope se directly nahi dikhta. [Q8, Q13, Q14]
- **Supernova** = dying star ka large explosion, jab star ke core mein change hota hai (2 types: binary system OR single star life-end). [Q15]
- Terms — **Event Horizon** (black hole boundary), **Singularity** (infinite gravity point), **String Theory**, **Standard Model** — sab Universe observation/understanding ke context mein aate hain. [Q28]

Astronomical Objects — Identification

- Astronomical objects: Pulsar, Black Hole, Quasar — sab space mein hote hain. [Q17, Q18]
- **Brittle Star** astronomical object NAHI hai — yeh ek marine echinoderm (Ophiuroidea class) hai, sea floor pe rehta hai, Starfish se related. [Q17, Q18]

⚠ EXAM TRAP

- Brittle Star naam mein 'Star' hai lekin yeh **sea creature** hai, astronomical object nahi — common confusion point. [Q17, Q18]

Distance Units — Light Year, AU

- **Light Year** = **distance** ki unit (na ki time) — light ek year mein vacuum mein jitni distance cover karta hai. [Q22, Q23]
- Tare-tare ke beech distance measure karne ke liye **Light Year** use hota hai. [Q23]

⚠ EXAM TRAP

- Light Year time ki NAHI, **distance** ki unit hai — common mistake. [Q22]

Space/Sky Observations — Equator, Pole Star, Directions

- Agar observer ko stars horizon ke perpendicular rise hote dikhein, toh wo **Equator** pe located hoga (celestial poles horizon pe lie karte hain equator se). [Q24]
- Pole Star north direction mein shine karta hai — East ki taraf jaane ke liye Pole Star ko apni **left side** pe rakhna chahiye. [Q25]

- **Saptarishi** (group of stars) pole/north direction batata hai. [Q26]
- Vacuum/space mein: sky **black** dikhta hai + stars **twinkle nahi** karte (atmosphere ke absence ki wajah se); background temp $\sim 2.7\text{K}$ hota hai jo Earth surface se bahut kam hai. [Q21]

■ Miscellaneous — Solar Eclipse trigger, Natural time units

- 'Byte' space terminology se related NAHI hai (computer term hai) — Telemetering, Weightlessness, Cislunar sab space terms hain. [Q27]
- Standard Time natural unit of time NAHI hai — Tropical year, Lunar month, Day sab natural units hain (Earth/Moon ki motion pe based). [Q29]